Anmol Goyal

Mathematics with Statistics @ King's College London · Visiting Student AI Researcher @ Oxford Dynamics

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Mathematics student and visiting AI researcher with hands-on experience in robotics and physical AI. Currently leading a final-year industry research project on **liquid neural networks** in collaboration with King's College London and Oxford Dynamics.

Education

King's College London — London, UK
BSc (Hons) Mathematics with Statistics — Sep 2024 – Jul 2027

- **Final year:** Informatics modules in Natural Language Processing and Computational Modelling, alongside core mathematics and statistics.
- **Prior modules:** Probability & Statistics II, Applied Probabilistic Models, Random Structures & Algorithms, Statistical Modelling, Real Analysis, Discrete Mathematics.

GD Goenka Public School, Gurugram, India · *High School Diploma in Science (Computer Science & Physics)* Apr 2011 – Mar 2024

Experience

Oxford Dynamics *(defence AI & robotics)* — Harwell Science Campus, Oxfordshire
Visiting Student AI Researcher — Sep 2026 – Present

- Stayed on after the internship to lead research on **liquid time-constant (LTC) networks as adaptive filters**, run as my KCL final-year project under joint academic and industry supervision.
- Investigating whether the learned, input-dependent time constants of LTC networks correspond to the **Kalman gain** of an optimal linear filter, connecting continuous-time neural ODEs with classical state estimation; built on open-source datasets so the work remains publishable.

Robotics Intern — Jun 2026 – Sep 2026

- Wrote **ROS 2 / MoveIt** motion-control scripts for a 6-DoF Universal Robots **UR7e** collaborative arm; the code runs on the hardware and is merged into the team repository.
- Helped commission the UR7e and built its **digital twin** in **NVIDIA Isaac Sim** for simulation-first testing of motion plans.
- Developed ROS 2 nodes and publisher–subscriber pipelines in containerised **Docker** environments, working to the team's branching and code-review practices.
- Reviewed the literature on AI foundations and world models to inform the research direction for learned models in autonomous systems, and on classical guidance and control, including null-seeking guidance.

Skills

Programming	Python (7 years, self-taught), SQL, R, JavaScript, Java, LaTeX
Robotics	ROS 2, MoveIt, NVIDIA Isaac Sim, Universal Robots (UR7e)
Data & ML	pandas, matplotlib, BeautifulSoup, statistical modelling (regression, diagnostics, model selection)
Mathematics	Probability, stochastic processes, state estimation (Kalman filtering), linear algebra
Tooling	Git, Docker, Linux, IBM Z & LinuxONE
Certifications	IBM Z Xplore: Concepts; IBM Z & LinuxONE Community Contributor 2025 (Level 1)
Languages	English (fluent), Hindi (native), Punjabi (intermediate), Spanish (elementary)

Leadership & Outreach

Co-President, IBM Z Society & IBM Z Student Ambassador, KCL — Jul 2025 – Jun 2026

- Represented IBM Z on campus: hosted talks with Amazon engineers and an IBM director, ran workshops, and grew a mainframe learning community on IBM's Z Xplore platform.

Founder & President, KCL Financial Literacy Society — Jan 2025 – May 2026

- Founded a society teaching students practical personal finance, from budgeting to investing.

NMES Faculty Student Representative, KCL — Oct 2025 – Sep 2026

- Selected by application to represent NMES students; analysed ∼150 survey responses and presented findings to staff.

Student Ambassador, KCL Mathematics (Subject Ambassador & Online Events) — Nov 2025 – Sep 2026

- Presented technical topics to audiences of 20–60 at outreach events and helped run online Q&As and webinars for applicants.

Participant, King's Entrepreneurship Institute Online Incubator — Aug 2025

Volunteer Python Instructor, Code Your Future (taught beginners in groups of 4–6) — Aug 2024 – Jan 2025